

Chapter 2 - Modules

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Exercise 1

Since $\gcd(m, n) = 1$, $\exists u, v \in \mathbb{Z}$, s.t.

$$mu + nv = 1$$

Now for $\forall x \otimes y \in (\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z})$,

$$\begin{aligned} x \otimes y &= (mu + nv)(x \otimes y) \\ &= mu(x \otimes y) + nv(x \otimes y) \\ &= 0 \end{aligned}$$

□

Exercise 2

Consider the exact sequence

$$\mathfrak{a} \xrightarrow{f} A \xrightarrow{g} A/\mathfrak{a} \rightarrow 0$$

By *Proposition 2.18*, we have the following exact sequence

$$\mathfrak{a} \otimes M \xrightarrow{f \otimes 1} A \otimes M \xrightarrow{g \otimes 1} (A/\mathfrak{a}) \otimes M \longrightarrow 0$$

Notice there exist trivial isomorphism $\mathfrak{a} \otimes M \cong \mathfrak{a}M$, $A \otimes M \cong M$. So we can have the exact sequence

$$\mathfrak{a}M \xrightarrow{\varphi} M \xrightarrow{\psi} (A/\mathfrak{a}) \otimes M \rightarrow 0$$

With $\text{Im } \varphi = \text{Ker } \psi$, $\text{Im } \psi = (A/\mathfrak{a}) \otimes M$. Which is to say

$$\text{Im } \psi = (A/\mathfrak{a}) \otimes M, \text{Ker } \psi = \mathfrak{a}M$$

So by *Module Isomorphism Theorem*,

$$(A/\mathfrak{a}) \otimes M \cong M/\mathfrak{a}M$$

□